

Radio Access and Link Control in Cellular Networks

Marceau Coupechoux

Telecom Paris, Institut Polytechnique de Paris
Département INFRES/RMS

Oct. 2025

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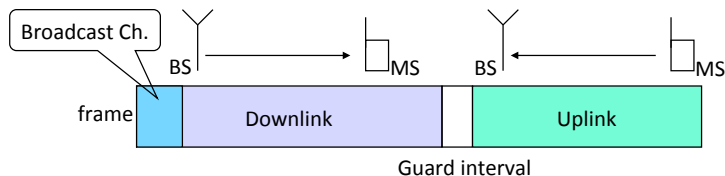
6 References

Duplexing schemes

- Uplink : communications from the Mobile Station (MS) to the Base Station (BS)
- Downlink : communications from BS to MS
- Duplexing : the way uplink and downlink are allocated to orthogonal radio resources
- Two main approaches : TDD and FDD.

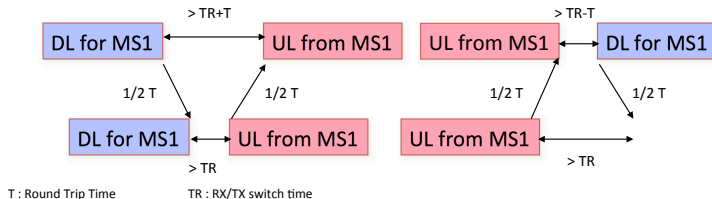
TDD I

- Time is divided in frames and every frame in two sub-frames
- UL and DL use the same carrier frequency during different sub-frames (e.g. in NR)
- Guard interval between UL and DL sub-frames to maintain orthogonality bw links despite propagation delays
- Synchronization is needed bw UL and DL



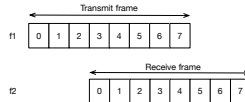
TDD II

- No need for stations to support full-duplex
- TDD systems can be deployed on non-peered spectrum bands
- Channels characteristics are similar on both links : open loop channel estimation is possible (and useful for MIMO, beamforming and power control)
- TDD is more adapted to small cells because of the small propagation time

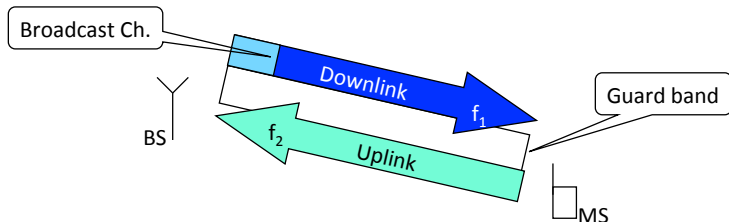


FDD

- UL and DL use different carrier frequencies
- In duplex mode, RX and TX can be simultaneous (e.g. in UMTS)
- In half-duplex, RX and TX cannot be simultaneous (e.g. in GSM)

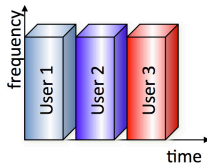
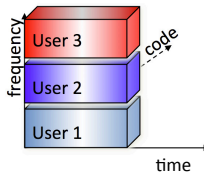
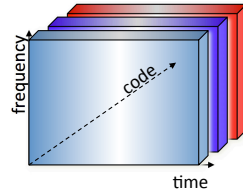


- Two peered frequency bands are usually needed
- Time synchronization is not needed in duplex mode
- Channel conditions on UL and DL are independent



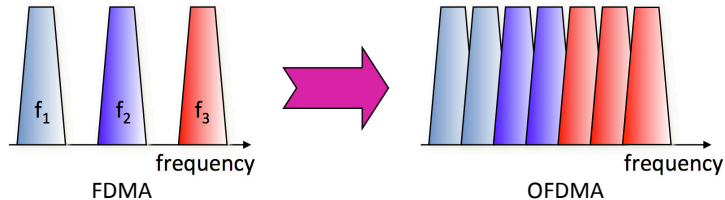
Multiplexing schemes

- Multiplexing schemes allocate different radio resources to different users sharing a common link (UL or DL)
- Signal space can be divided in time, frequency or codes to obtain TDMA, FDMA or CDMA (Time/Frequency/Code Multiple Access)

**TDMA****FDMA****CDMA**

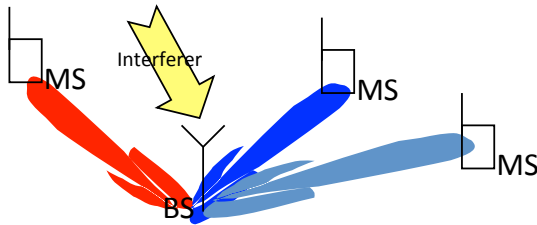
OFDMA

- OFDMA (Orthogonal Frequency Division Multiple Access) can be seen as a special case of FDMA
- Every elementary frequency band is called a *sub-carrier*
- Main advantages : ease of implementation, better spectral efficiency, easy equalization, flexible resource allocation
- Main drawbacks : high peak to average power ratio, inter-carrier interference



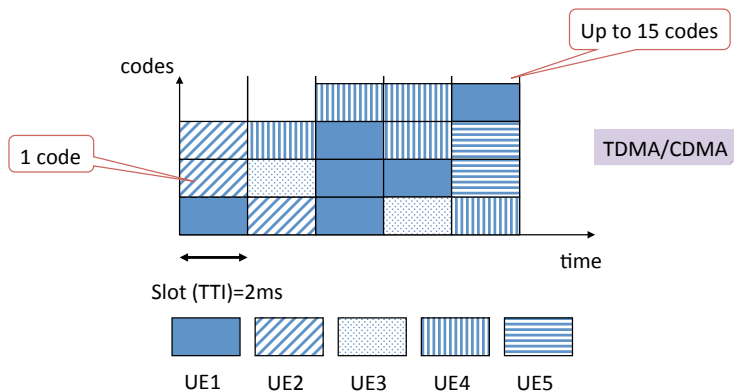
SDMA

- The use of beamforming or Multi-user MIMO adds a new dimension : Space Division Multiple Access (SDMA)



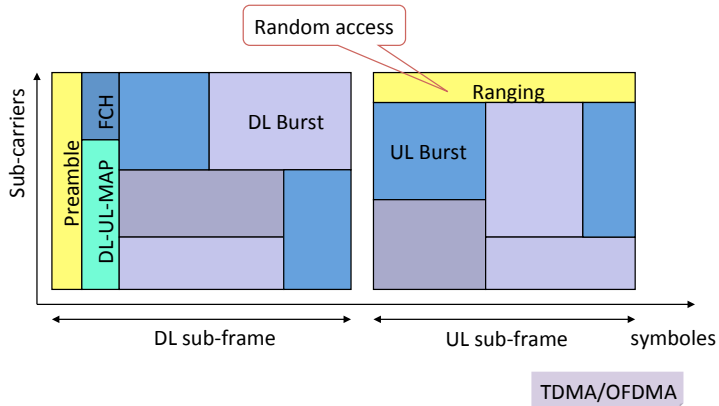
Examples I

- In HSPA, a data channel is slotted and shared by all DL User Equipments (UE). Several UEs can receive data on the same slot (or Transmission Time Interval – TTI) but encoded with different orthogonal codes



Examples II

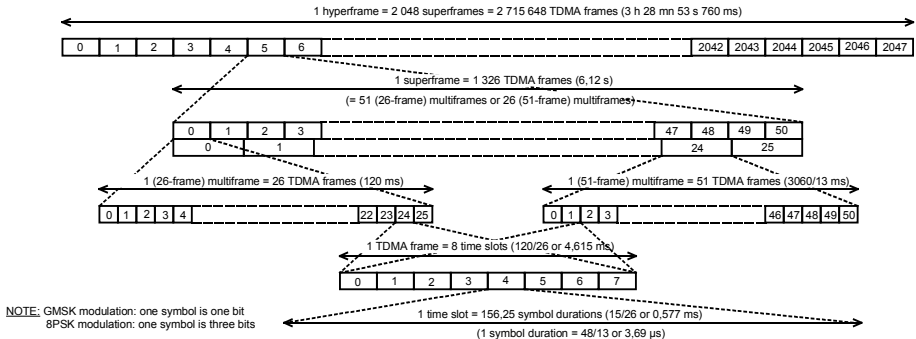
- In WiMAX there are two degrees of freedom for multiplexing : sub-carriers (of an OFDM symbol) and time (OFDM symbols)
- Duplexing scheme is TDD



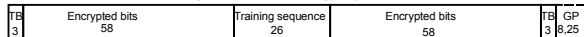
Examples III

- In GSM, operator spectrum is divided in 200 KHz channels, each frequency channel is divided in 8 slots. Duplexing scheme is FDD.
- Frames are grouped together to form multiframes (26 or 51 frames). Multiframes are grouped in superframes and super frames in hyperframes.

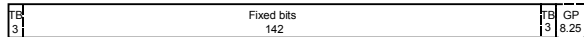
Examples IV



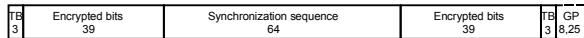
Normal burst (NB)
The number shown are in symbols



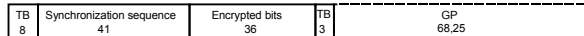
Frequency correction burst (FB)



Synchronization burst (SB)

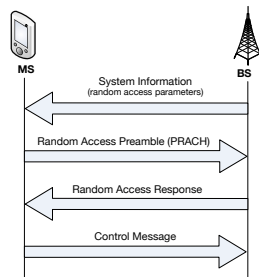


Access burst (AB)



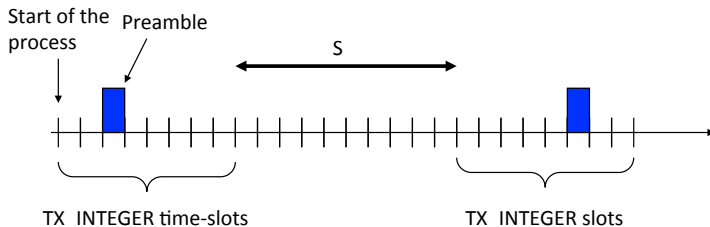
Random access procedure

- 1 the MS acquires random access parameters from the broadcast channel (position of the PRACH, protocol parameters, transmit power, etc)
- 2 the MS transmits a short *burst* or *preamble* using slotted Aloha. Transmissions are subject to collisions and capture.
- 3 in case of collision, the MS waits for a random duration (back off) and retransmits the preamble
- 4 the BS sends a random access response that includes at least the timing advance and an uplink allocation
- 5 the MS sends a control message on the allocated resource with at least the reason for performing random access (mobile originated call, response to a paging, location update, etc)



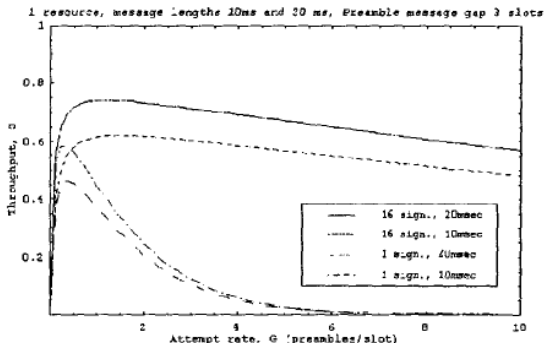
Random access protocol I

- All random access protocols are based on slotted ALOHA
- Example of GSM : TX_INTEGER is the back-off window (e.g. bw 8 and 50 slots), S is the minimum delay before retransmissions (e.g. 250 ms), there is a maximum of MAX_TRANS retransmissions
- All parameters are broadcast by the Broadcast Control Channel (in a System Information Block)
- In GSM : the preamble includes a service class and a random number to cope with the capture effect



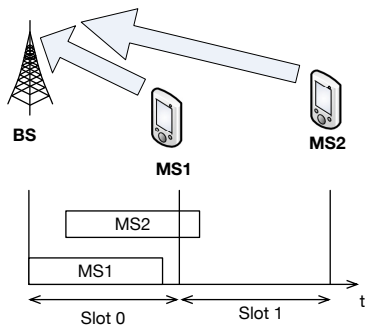
Random access protocol II

- In UMTS and in LTE, the User Equipment (UE) chooses also a random signature (a scrambling code in UMTS, a Zadoff-Chu sequence in LTE)
- UMTS example (source : I. N. Vukovic and T. Brown, "Performance Analysis of the Random Access Channel in WCDMA", IEEE VTC'01) :



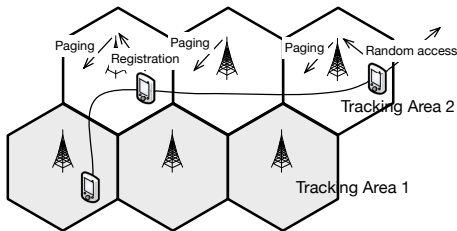
Timing Advance

- **Goal** : synchronization of UL signals in order to maintain orthogonality between MSs of a cell
- After random access, the BS transmits a timing offset for all subsequent uplink transmissions of the MS. Timing advance is regularly updated.

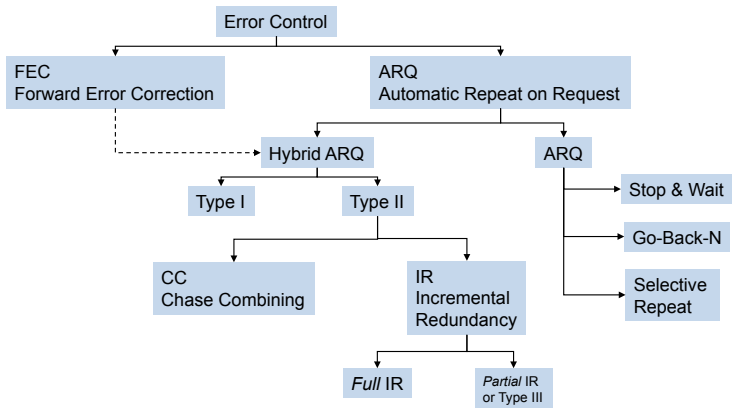


Paging

- **Goal :** Paging is used for incoming calls or SMS.
- Paging messages include TMSIs of the users who need to be connected.
- They are sent by all BS of the location/tracking area.
- Upon reception of a paging message destined to it, the user performs a random access, which includes the reason of the access (here : paging response).
- Discontinuous reception : based on their TMSIs, users decode only a subset of paging messages.

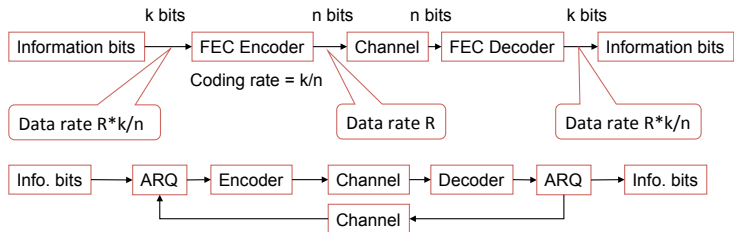


Classification



FEC vs ARQ

- FEC : the transmitter adds redundant information to the useful information and the receiver exploits this redundancy to decode the information bits. E.g. : convolutional codes, LDPC, turbo codes, etc. No need for a feedback channel. There is a data rate vs error correction power trade-off.
- ARQ : based on an error detection code, the receiver sends back an ACK/NACK upon reception of a packet (NB : ARQ can be also implemented at layer 4, cf TCP)

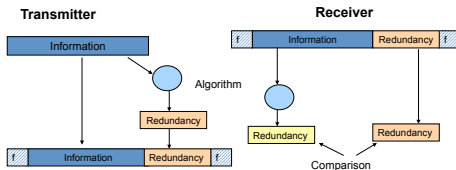


Error detection

- For every ARQ protocol, there is a need for an error detection scheme (source of the figure : [1])
- Cyclic Redundancy Codes (CRC) are usually used for error detection with very good performances (see [18])
- For a rate k/n , a bound on the probability to not detect an error is :

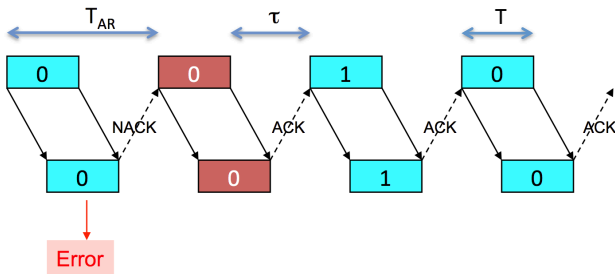
$$P_e < 2^{k-n}$$

- Proof : If the bit error probability is $\epsilon = 1/2$, the 2^n n -tuples that are likely to be received with the same probability 2^{-n} . Among these tuples, $2^k - 1$ are recognized as code words and still different from the transmitted code words. We thus have : $P_e = (2^k - 1)2^{-n} = 2^{k-n} - 2^{-n} < 2^{k-n}$.



Stop and Wait I

- Frames are numbered modulo 2
- Let n be the number of bits, R the data rate and τ the idle time
- Let $T = \frac{n}{R}$ be the transmission time
- Let $T_{AR} = T + \tau$ be the round trip time.



Stop and Wait II

- Consider the Binary Symmetric Channel (BSC) with error probability ϵ
- Let P_c be the probability that a n bit frame has no error.
- Let P_d be the probability that a n bit frame has errors that **can** be detected by the receiver.
- Let P_e be the probability that a n bit frame has errors that **cannot** be detected by the receiver.

$$P_c + P_d + P_e = 1$$

- Let R be the data rate.

Stop and Wait III

- The probability to deliver to higher layers a frame with undetected errors is very small :

$$P(E) = \sum_{i=0}^{\infty} P_e P_d^i = \frac{P_e}{1 - P_d}.$$

It is supposed to be negligible.

- Assume error on bits are independent, we have :

$$P_c = (1 - \epsilon)^n$$

- We have a bound for P_e :

$$P_e < 2^{k-n}$$

- Consider a BCH code with $k = 2014$ and $n = 34$, then $P(E) = 8.10^{-10}$.

Stop and Wait IV

- Average delay to deliver a frame with Stop and Wait : (assuming P_e is negligible)

$$\delta_{SW} = \sum_{i=1}^{\infty} iT_{AR}P_c(1-P_c)^{i-1} = \frac{T_{AR}}{P_c} = \frac{\frac{n}{R} + \tau}{1 - P_d}$$

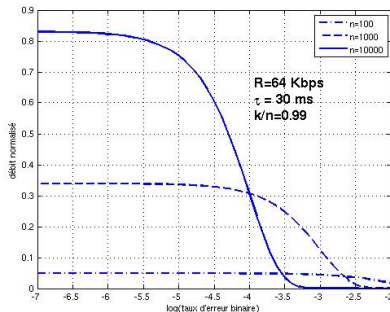
- Goodput of Stop and Wait :

$$\eta_{SW} = \frac{k}{\delta_{SW}} = \frac{1}{1 + \frac{\tau R}{n}} \frac{k}{n} R(1 - P_d)$$

- Example : $R = 1$ Mbps, $\tau = 700$ ms (satellite link), $\tau R = 7 \cdot 10^5$. In order that $\tau R/n$ be close to 1, we need $n \approx 10^6$ bits. With $n = 10000$ bits, we have already $\tau R/n = 70$.

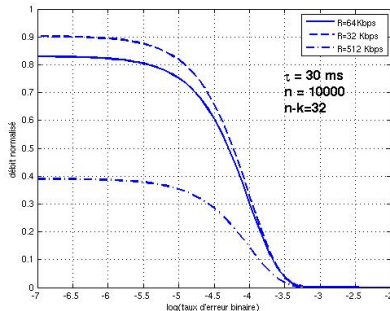
Stop and Wait V

- Goodput as a function of the (log of the) bit error rate
- When n increases, τ becomes small in front of $T = n/R$.



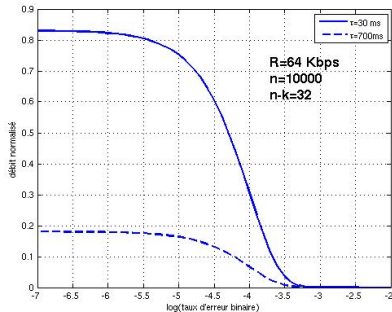
Stop and Wait VI

- When R increases, T becomes small in front of τ .



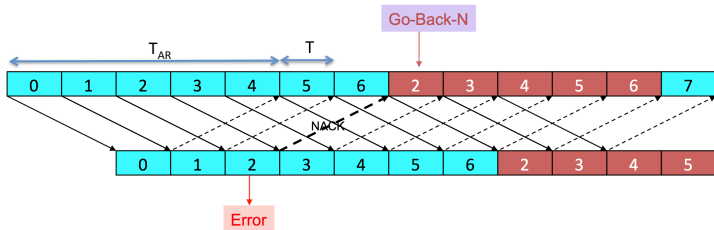
Stop and Wait VII

- When τ increases, idle time increases.



Go Back N I

- Anticipation window : the transmitter is allowed to transmit up to W frames without receiving corresponding ACKs.
- For a window of W frames, frames have to be numbered modulo $W + 1$.
- We assume below that the window size is ideal, i.e., $W = T_{AR}/T$.



Go Back N II

- Average delay to deliver a frame with Go Back N :

$$\begin{aligned}
 \delta_{GBN} &= \sum_{i=1}^{\infty} T((i-1)W + 1)P_c(1 - P_c)^{i-1} \\
 &= TP_c(W \sum_{i=1}^{\infty} i(1 - P_c)^i + \sum_{i=1}^{\infty} (1 - P_c)^{i-1}) \\
 &= TP_c \left(\frac{W(1 - P_c)}{P_c^2} + \frac{1}{P_c} \right) \\
 &= T \left(1 + W \frac{1 - P_c}{P_c} \right)
 \end{aligned}$$

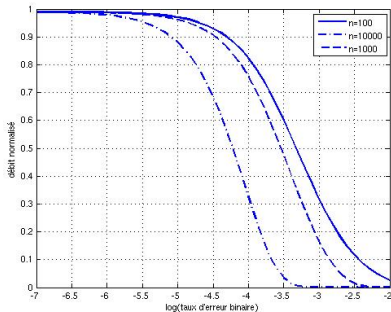
- Goodput of Go back N :

$$\eta_{GBN} = \frac{1}{1 + P_d(W - 1)} \frac{k}{n} R(1 - P_d)$$

- If R and τ are small, W can be set to a small value and the good put of Go Back N is good. If R is high and the propagation time is long, the denominator is high and the goodput is low.

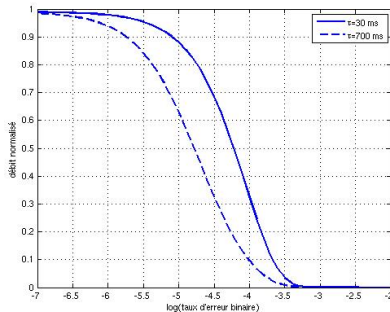
Go Back N III

- When n increases, P_d and W increase.



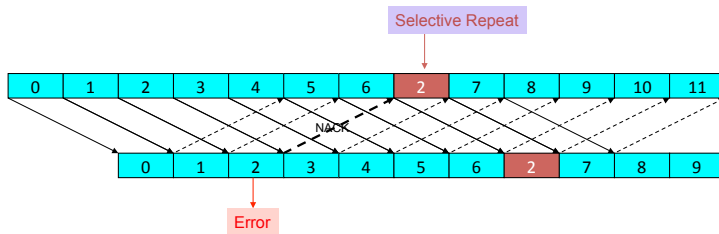
Go Back N IV

- When τ increases, W increases.



Selective Repeat I

- The transmitter retransmits only the frames for which it received a NACK
- For a window of W frames, frames have to be numbered modulo $2W$.



Selective Repeat II

- Average delay to deliver a frame with Selective Repeat :

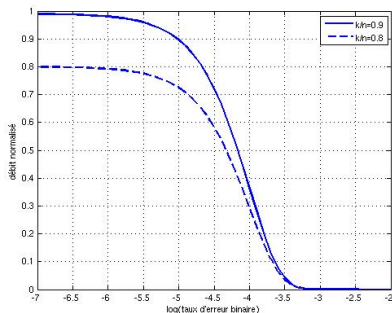
$$\begin{aligned}\delta_{SR} &= \sum_{i=1}^{\infty} iTP_c(1 - P_c)^{i-1} \\ &= \frac{T}{P_c}\end{aligned}\tag{1}$$

- Goodput of Selective Repeat :

$$\eta_{SR} = \frac{k}{n}R(1 - P_d)$$

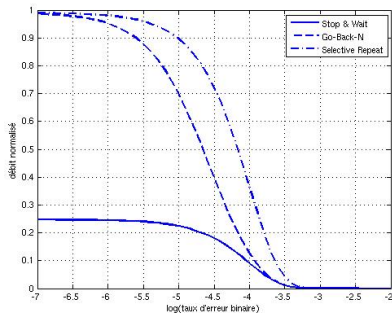
Selective Repeat III

- Impact of k/n .



Selective Repeat IV

- Comparison in terms of goodput.

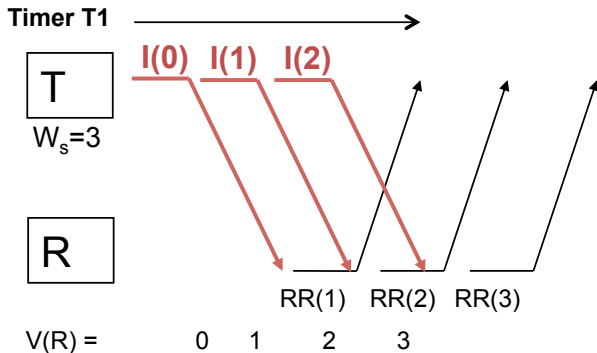


Example : LAP-B I

- Used in X25 but very close to LAP-Dm (used in GSM for control channels) and UMTS/LTE/NR RLC Acknowledge Mode
- Every frame has a sequence number indicated in its header, numbering is done modulo $M \in \{2, 128, 32768\}$
- A connection-oriented protocol : transmitter and receiver sides maintain counters that are initialized when connection is open. After transmissions, connection is closed
- Go-Back-N : the transmitter can send up to W_s frames without receiving corresponding ACKs
- Ready to Receive (RR) : may acknowledge several frames
- After a Reject (REJ) : all frames from the requested SN are retransmitted
- The receiver maintains a counter of the next SN to be received $V(R)$

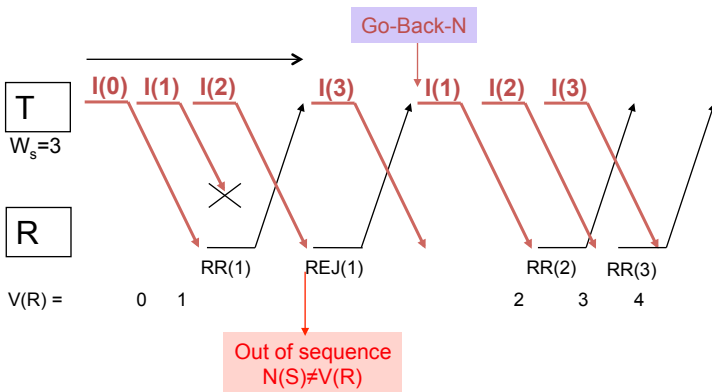
Example : LAP-B II

- Information frames (I) includes $N(S)$ = its sequence number
- RR includes $N(R)$ = sequence number of the next expected frame
- $V(R)$ = receiver counter indicating the next expected frame
- Every transmission is associated to a timer T1



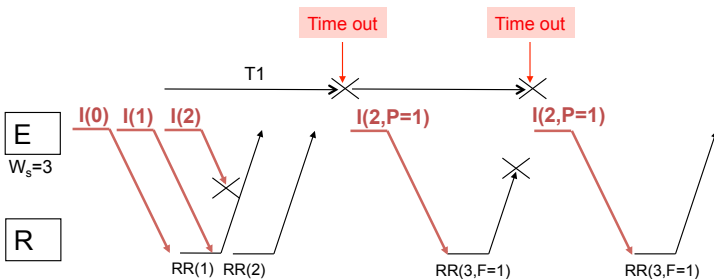
Example : LAP-B III

- Retransmission upon negative acknowledgement (REJ)



Example : LAP-B IV

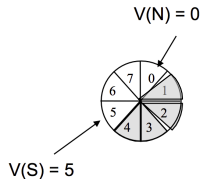
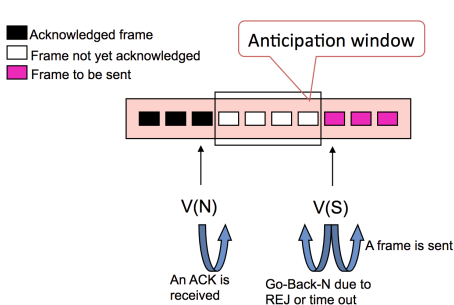
- Retransmission upon T1 time out
- P/F bit : an immediate response is required / immediate response
- There is a maximum number of retransmissions



Example : LAP-B V

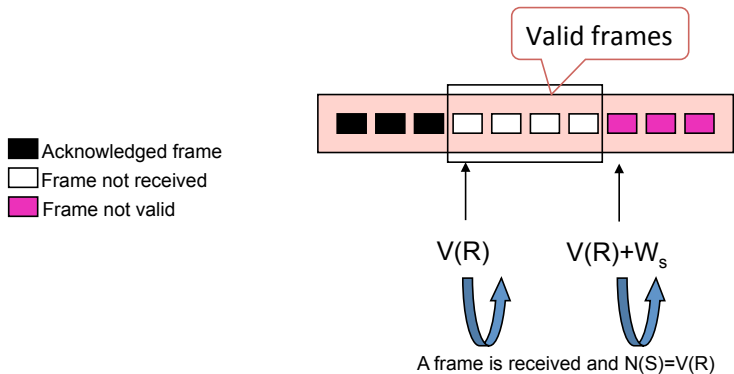
• Transmitter side :

- $V(N)$ = last acknowledged frame
- $V(S)$ = next frame to send
- NB : $V(S) - V(N) - 1 \leq W_s$



Example : LAP-B VI

- Receiver side :
 - $V(R)$ = next expected frame

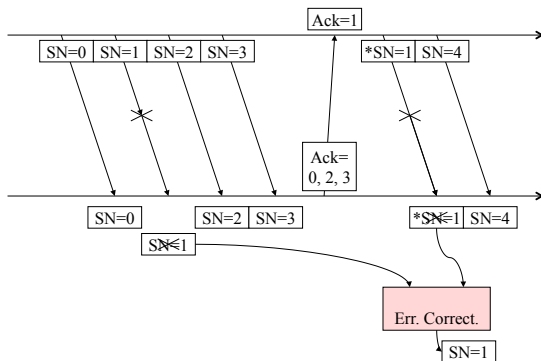


HARQ Types

- Type I : FEC and ARQ are well separated
- Type II : the receiver keeps a copy of packets in error and use the information to decode subsequent packets
- Type II with Chase Combining (CC) : all retransmissions are identical
- Type II with Incremental Redundancy (IR) : the coding scheme is modified at each retransmission of the same packet
- Type II Partial IR : every transmission includes information bits + parity bits
- Type II Full IR : the first transmission includes information bits + parity bits, retransmissions includes only parity bits

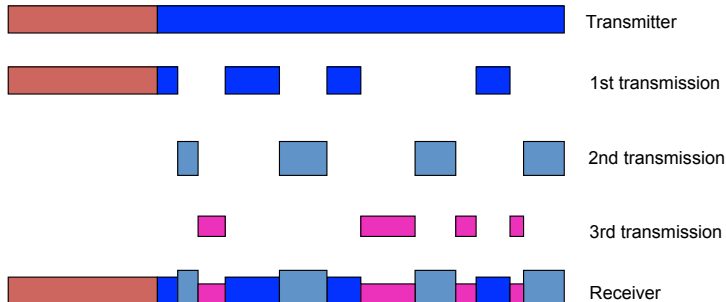
HARQ Type II Illustrated

- On this example, we can obtain a correct copy of packet with Sequence Number SN=1 from two erroneous receptions
- With CC, the two transmissions of (SN=1) are identical. With IR, the second transmission is re-encoded



HARQ Full IR Illustrated

- In IR, information is first coded with a low rate code at the transmitter. Information bits and some parity bits are sent at the first transmission
- If a NACK is received, the transmitter sends additional parity bits. The receiver aggregates received bits to those already received
- At every reception, the effective coding rate is decreased
- In Full IR, information is sent only at the first transmission. In Partial IR, information is sent at every transmission



Chase Combining

To go in more details...

- **Main idea** : if the initial FEC coding rate is R , then after L transmissions, the effective rate is R/L
- Let $Y_i = (Y_{i,1}, Y_{i,2}, \dots, Y_{i,N})$ be the i -th copy of the packet (made of N bits)
- After the second transmission, transmitted bits can be written :

$$(Y_{1,1}, Y_{2,1}, Y_{1,2}, Y_{2,2}, \dots, Y_{1,N}, Y_{2,N})$$

which can be seen as the superposition of the initial FEC and a repetition code. There are $N \times R$ information bits and the packet length is $2N$, so that the new code rate is $NR/(2N) = R/2$

- Decoding can be done using maximum likelihood and Viterbi algorithm
- Maximum likelihood : we look for the sequence s that maximizes $P[s|r]$, where r is the received sequence

Incremental Redundancy I

To go in more details...

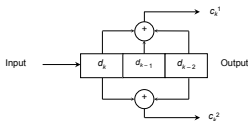
- **Main idea** : IR is based on a sequence of codes obtained by increasingly puncturing a mother code
- **Puncturing** : remove periodically some bits from the encoder output (increases code rate and decreases decoding complexity)
- **Example** : Take a code with rate $R_1 = 1/2$:

$$C_1 = (c_0(1), c_0(2), c_1(1), c_1(2), \dots)$$

We obtain a code of rate $R_2 = 2/3$ by puncturing one bit out of 4 :

$$C_2 = (c_0(1), c_0(2), c_1(1), -, c_2(1), c_2(2), c_3(1), -, \dots)$$

We obtain a code of rate $R_3 = 4/5$ by puncturing 3 bits out of 8, etc



Incremental Redundancy II

- Rate Compatible Punctured Convolutional Codes (RCPC) :

- A family of codes $\{C_n\}$ is obtained by puncturing a low rate mother code with an increasing number of bits, so that $R_n \leq R_{n+1}$
- Punctured bits to obtain C_n are included in the set of punctured bits to obtain C_{n+1}

- Example :

0 bit punctured (mother code) :

$$C_1 = (c_0(1), c_0(2), c_1(1), c_1(2), \dots), \quad R_1 = 1/2$$

1 bit punctured out of 8 :

$$C_2 = (c_0(1), c_0(2), c_1(1), c_1(2), c_2(1), -, c_3(1), c_3(2), \dots), \quad R_2 = 4/7$$

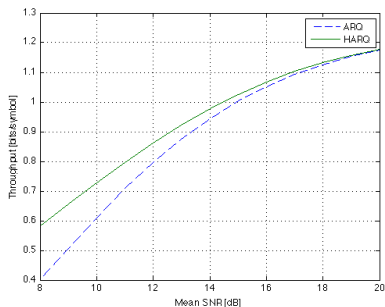
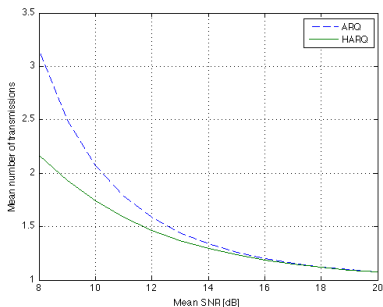
2 bits punctured out of 8 :

$$C_3 = (c_0(1), c_0(2), c_1(1), c_1(2), c_2(1), -, -, c_3(2), \dots), \quad R_3 = 4/6$$

- C_n offers a higher protection against errors than C_{n+1} , however with a lower useful data rate

HARQ vs ARQ

- Performance comparison based on [30] (MCS4, 16-QAM, rate=9/16, 2.25 bits/symbol, $g = 0.664$, $\gamma_M = 7.7$ dB)



Open Loop Power Control

- Open loop power control : based on BS transmit power ($P_{tx_{BS}}$), target received power (P_{ra}) and received power (P_{rx}), the UE sets its transmit power for random access ($P_{tx_{UE}}$)

Path-loss estimation on the downlink :

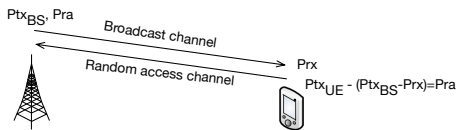
$$PL = P_{tx_{BS}} - P_{rx} \quad (2)$$

Estimation of the power received at the BS on the uplink :

$$P_{rx_{BS}} = P_{tx_{UE}} - PL \quad (3)$$

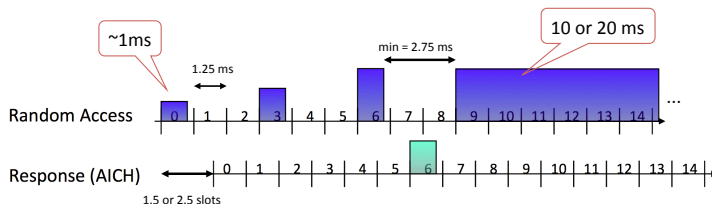
When $P_{rx_{BS}} = P_{ra}$:

$$P_{tx_{UE}} = P_{ra} + PL \quad (4)$$



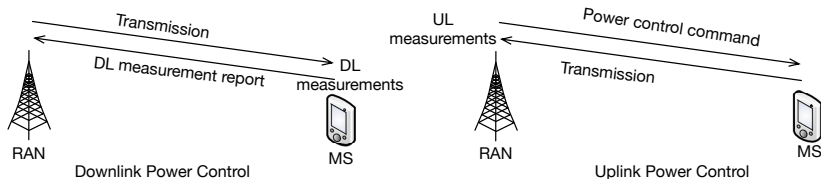
Open Loop Power Control

- Power ramping : at each retransmission (i.e. if there is no positive feedback), the transmit power is increased (typically +1 dB) up to some maximum value. This approach aims at compensating potential mis-estimations of the transmit power. Example of UMTS :



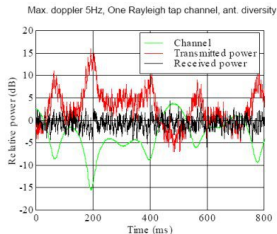
Closed Loop Power control

- DL : based on DL transmissions, the MS makes DL measurements (signal level and quality) and feedbacks them in a measurement report. The RAN adjusts its transmit power
- UL : based on UL transmissions, the RAN performs UL measurements and sends power control command to the MS, which adjusts its power



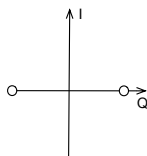
Slow and Fast Power Control

- Slow power control follows the slow variations of the radio channel (due to shadowing). Example in GSM :
 - Measurements reports and PC commands are sent every 480 ms on the SACCH
 - Upon reception of a command, a MS changes its transmit power by steps of 2 dB every 60 ms until reaching the target transmit power
- Fast power control follows the fast variations of the channel (due to fast fading). Example in UMTS : Power control rate is every $666 \mu\text{s}$ (1500 Hz)

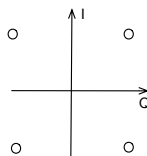


Modulations

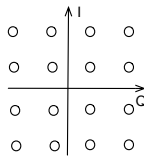
- Traditional modulations used in cellular networks : BPSK, QPSK, GMSK (GSM), 16-QAM (HSPA), 64-QAM (LTE), 256-QAM (NR)
- In “X-QAM”, X represents the number of QAM symbols in the constellation. The number of bits per QAM symbol is $\log_2(X)$. For example, in 64-QAM, there are $\log_2(64) = 6$ bits per QAM symbol.
- For a fixed symbol duration T_s , the raw data rate (i.e. ignoring channel coding) is $R = \log_2(X)/T_s$. The denser the modulation (high X), the higher the raw data rate.



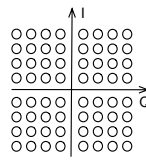
BPSK (1 bit/symbol)



QPSK (2 bits/symbol)



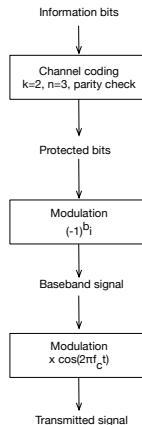
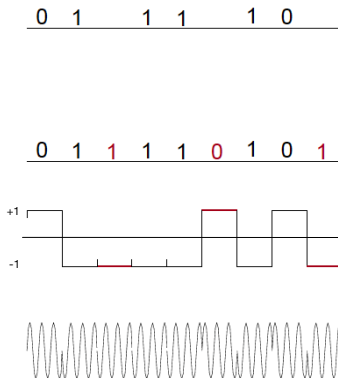
16-QAM (4 bits/symbol)



64-QAM (6 bits/symbol)

Modulations

Example of BPSK

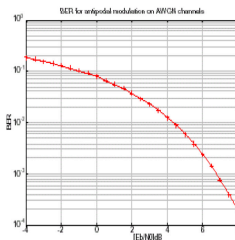
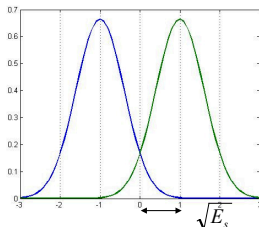


Modulations

- However : denser is the modulation, the more sensitive it is to noise
- Example : in BPSK, there is one bit per symbol $s \in \{-1, +1\}$.
- Consider the event A_{01} : -1 is sent but the received signal $r = \sqrt{E_s}s + w$ is closer to $+\sqrt{E_s}$.

$$\mathbb{P}[A_{01}] = \mathbb{P}[w > \sqrt{E_s}] = Q\left(\sqrt{\frac{2E_s}{N_0}}\right) \quad (5)$$

where E_s is the received signal energy per symbol (i.e. bit here). $N_0/2$ is the variance of AWGN.



Modulation and Coding Scheme

Recall that

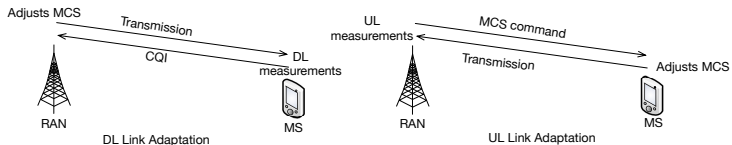
- A FEC encoder takes k bits as input and outputs $n \geq k$ bits. Coding rate is $r = \frac{k}{n}$. For a data rate R , information rate is rR
- When $r \rightarrow 1$: few redundancy bits, less error correction, high information rate.
When $r \rightarrow 0$: many redundancy bits, more error correction, low information rate.

MCS :

- A combination of a modulation and a coding rate
- Example : MCS=(64-QAM, $r = 1/3$). With 64-QAM, we transmit 6 bits per symbol (every T_s), but 2 bits over 3 are added by the FEC encoder. The information data rate is thus $6 \times 1/3 = 2$ bits per symbol duration.
- When the channel is bad, robust modulations and low coding rates are favoured.
When the channel is good, dense modulations and higher coding rates are favoured.

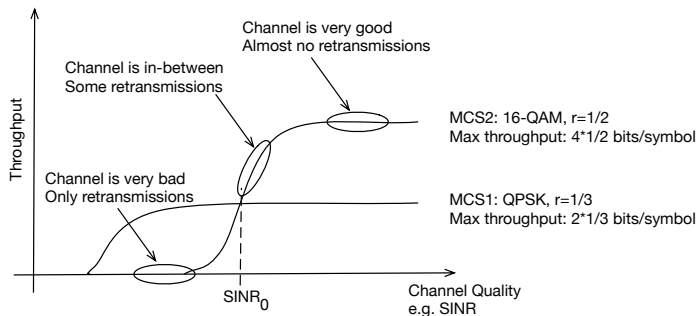
Link Adaptation

- An algorithm that dynamically adjusts modulation and coding rate according to the radio channel variations
- Adjustment is based on measurements. DL : the MS makes channel measurements and feeds back a Channel Quality Indicator (CQI), the network adjusts the Modulation and Coding Scheme (MCS). UL : the network makes measurements and sends a MCS command to the MS, which adjusts its transmission
- CQI : indicates the MCS providing the highest data rate with a Block Error Rate (BLER) less than 10%



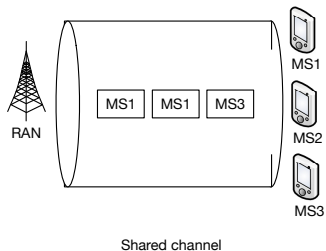
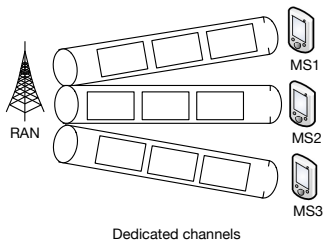
Example of Algorithm

- If $\text{SINR} \leq \text{SINR}_0$, then choose MCS1
- If $\text{SINR} > \text{SINR}_0$, then choose MCS2



Dedicated vs. Shared Channel

- Dedicated channel (GSM, UMTS R'99) : the traffic of a user is transmitted over dedicated resources (slot, frequency, code) allocated for the whole connection duration. No need to address frames. Channel reconfiguration may be performed by RRC.
- Shared channel (HSPA, LTE, NR) : the whole user traffic is transmitted over a single channel. Allocation is done by the scheduling algorithm at a very short time scale. Every frame includes an address.



Downlink scheduling (one-to-many)

- One-to-many for the DL : scheduling information is available at the scheduler, CQI is fed back by MSs

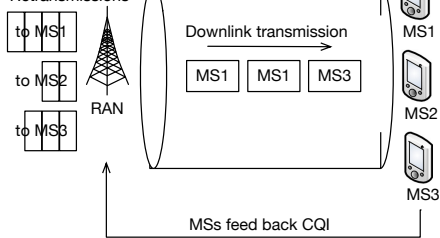
Scheduling Information:

Buffer status

Priorities

QoS requirements

Retransmissions



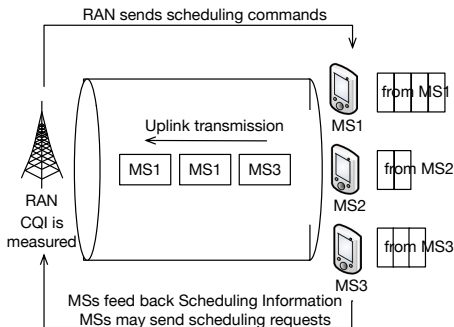
One-to-many scheduling

Scheduling information :

- Buffer status : number of packets in the buffers
- Priorities : between channels and services
- QoS requirements : data rate, latency
- Retransmissions : fresh packet or retransmission

Uplink scheduling (many-to-one)

- Many-to-one for the UL : scheduling is still controlled by the network through scheduling commands. Scheduling information is fed back from the MSs. CQI is measured on UL signals. If a MS is not known from the scheduler, this MS can send some scheduling request.

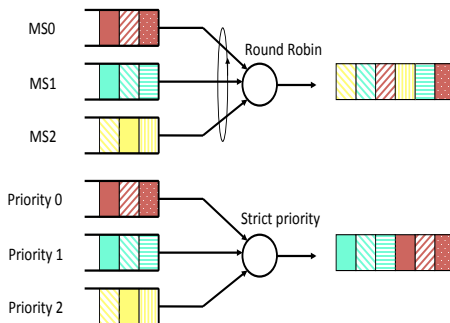


Many-to-one scheduling

Examples of Scheduling Algorithms

Basic approaches :

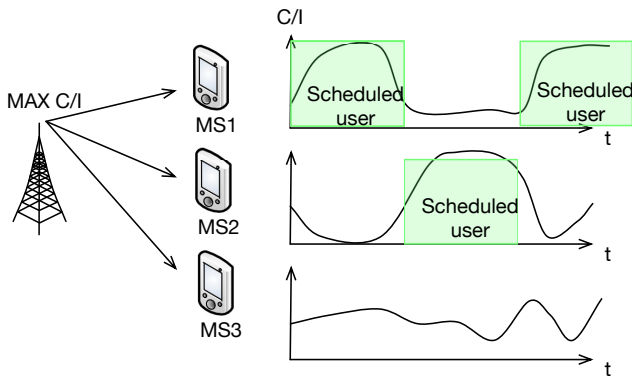
- Round Robin : every user is served in turn. If all slots are equal, Round Robin is fair in radio resource, i.e., all users get the same share of the bandwidth.
- Strict priority : as long as there is traffic for a higher priority queue, it is served. If there is no packet with priority 0 to be served, packets with priority 1 are served, etc.



Examples of Scheduling Algorithms

Channel aware schedulers :

- Max C/I : the MS with the best instantaneous channel is served
- Proportional fair scheduler : the scheduler allocated the resource to the MS with the best $\frac{r_i(t)}{\bar{R}_i}$ ratio, where $r_i(t)$ is the instantaneous data rate and $\bar{R}_i$ is the average throughput



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